Boston University, Department of Computing & Data Sciences

M.S. Thesis Proposal

What Does a Physics Network Actually Learn? Mechanistic Interpretability Across a Spectrum of Physics-Informed Neural Architectures

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Problem Statement

Physics-informed neural networks encode physical knowledge in two fundamentally different ways. Some architectures are *symmetry-constrained*: they are designed so that outputs obey known symmetries (e.g., rotational equivariance) regardless of what the network has learned internally. Others are *physics-explicit*: they directly parameterize a physical quantity such as an energy or Hamiltonian, enforcing conservation laws through the structure of the computation itself. Both approaches achieve strong empirical performance on physical dynamics tasks. But a foundational question has not been answered:

Do physics-explicit and symmetry-constrained architectures develop the same internal representations when trained on the same physical system, or do they discover fundamentally different computational strategies to produce the right answer?

More concretely: when a network trained on Coulomb dynamics produces accurate predictions, has it learned Coulomb's law, or has it found a different pattern that happens to generalize within the training distribution? Symmetry constraints guarantee that outputs transform correctly under rotation. Physics-explicit structure guarantees energy conservation. Neither guarantee tells us

whether the network's hidden representations correspond to physically meaningful quantities (forces, charge interactions, distance-dependent potentials).

This thesis applies mechanistic interpretability methods to directly examine, compare, and characterize the internal representations of five neural architectures spanning a spectrum from physics-explicit to fully unconstrained, all trained on an identical Coulomb particle dynamics task where ground-truth physical quantities are analytically computable at every point.

Background

A Spectrum of Physics Priors

Modern physics-informed neural networks occupy a spectrum based on how deeply physical knowledge is encoded:

Tier	Architecture	Physics Prior
Physics-explicit	Hamiltonian Neural Network	Learns energy $H(q, p)$ directly; conservation enforced by construction
Symmetry-constrained	EGNN	Equivariant message passing
	Attention EGNN	Equivariant + learned interaction weights
	Equivariant Transformer	Equivariant attention (Equiformer)
Unconstrained	Non-equivariant GNN	Permutation symmetry only
	MLP	No structural prior

Equivariant Graph Neural Networks

The EGNN[1] performs message passing with coordinate updates constructed from equivariant operations on relative position vectors, ensuring $f(Rx, Rv) = R \cdot f(x, v)$ for all $R \in \mathrm{SO}(3)$. Each layer computes:

$$m_{ij} = \phi_e(h_i,\, h_j,\, \|x_i - x_j\|^2,\, a_{ij}) \tag{1}$$

$$x_i' = x_i + \sum_{j \neq i} (x_i - x_j)\, \phi_x(m_{ij}) \tag{2}$$

$$h_i' = \phi_h\Big(h_i,\, \textstyle\sum_{j \neq i} m_{ij}\Big) \tag{3}$$

where h_i are node features, x_i positions, and ϕ_e, ϕ_x, ϕ_h learned MLPs.

Hamiltonian Neural Networks

Hamiltonian Neural Networks (HNNs)[2] learn a scalar function $\mathcal{H}_\theta(q, p)$ and define dynamics via Hamilton's equations:

$$\dot{q} = \frac{\partial \mathcal{H}_\theta}{\partial p}, \qquad \dot{p} = -\frac{\partial \mathcal{H}_\theta}{\partial q}$$

The loss penalizes violations of these equations, so the network is explicitly trained to model conserved energy. For Coulomb dynamics, a well-trained HNN should recover $\mathcal{H} = \sum_i \frac{p_i^2}{2m} + \sum_{i<j} \frac{q_i q_j}{\|x_i - x_j\|}$.

Equivariant Transformers

Equiformer[3] replaces message passing with equivariant attention over irreducible representations of SO(3), achieving the same symmetry guarantees as EGNN through a different computational mechanism. This enables a direct test of whether symmetry-constrained architectures converge to similar internal representations regardless of their specific implementation.

Mechanistic Interpretability

Mechanistic interpretability[4;5] aims to reverse-engineer the algorithms neural networks implement. This thesis uses four complementary tools:

- **Linear probing:** Train lightweight classifiers on internal activations to test whether specific physical quantities are linearly encoded at each layer.
- **Causal intervention / activation patching:** Corrupt a specific input (e.g., flip charge q_i) and trace which components causally carry that change to the output.
- **Probing + ablation loop:** After identifying a probed feature (e.g., distance encoded at layer 2), ablate that direction in representation space and measure the resulting performance drop, upgrading from "the model encodes X" to "the model *uses* X to make predictions."
- **Representational similarity (CKA):** Compare the geometry of internal representations across architectures to test whether different models develop structurally equivalent solutions.

The Coulomb Dynamics Testbed

N particles with charges $q_i \in \{-1, +1\}$ interact via Coulomb forces:

$$F_{ij} = q_i q_j \frac{x_i - x_j}{\|x_i - x_j\|^3}$$

Given initial positions and velocities, the task is to predict positions after $T = 1000$ integration steps. The governing law is fully known, providing ground-truth physical quantities (pairwise distances, charge products, force magnitudes and directions, total energy) against which internal representations can be directly evaluated. This is the key advantage of this testbed over molecular or protein datasets: there is no ambiguity about what the model "should" be computing.

Proposed Research

Central Thesis Claim

We hypothesize that physics-explicit inductive bias (HNN) and symmetry-based inductive bias (EGNN, Equiformer) both improve generalization, but produce *fundamentally different* internal representations. Specifically: only physics-explicit architectures will develop hidden representations that directly correspond to physical law (i.e., probes for Coulomb potential and energy will succeed at intermediate layers), while symmetry-constrained architectures will encode physically interpretable geometry (distances, relative positions) without necessarily encoding the force law itself.

Research Questions

1. Do internal representations encode physically interpretable quantities (pairwise distance, charge-product sign, force magnitude, force direction, potential energy), and at which layers do these emerge across architectures?
2. Are probed representations causally active (does ablating a probed direction degrade performance, or is it a correlate rather than a cause?)
3. Do equivariant architectures with different mechanisms (EGNN vs. Equiformer) converge to structurally similar representations (via CKA), or does the implementation of symmetry matter?
4. Does the HNN's explicit energy parameterization produce representations more aligned with Coulomb physics than symmetry-constrained models, despite comparable output accuracy?
5. Do any architectures exhibit grokking (a delayed generalization transition), and if so, does

the emergence of physics-aligned representations coincide with this transition?

Architecture Suite

Five architectures will be trained on the identical Coulomb dynamics dataset under matched hyperparameter conditions:

Architecture	**Tier**	**Equivariant?**	**Key Question**
Hamiltonian NN	Physics-explicit	Partial	Does it learn H?
EGNN	Symmetry-constrained	Yes	Baseline physics GNN
Attention EGNN	Symmetry-constrained	Yes	Does attention learn $1/r^2$?
Equivariant Transformer	Symmetry-constrained	Yes	Same symmetry, different mechanism
Non-equivariant GNN	Unconstrained	No	What breaks without symmetry?

Experimental Plan

Phase 1: Probing for Physical Quantities

For each architecture, extract hidden activations at every layer and train linear probes to predict: (a) pairwise Euclidean distance $\|x_i - x_j\|$, (b) charge product $q_i q_j \in \{-1, +1\}$, (c) force magnitude $\|F_{ij}\|$, (d) force direction $\hat{F}_{ij}$, (e) total potential energy $U = \sum_{i<j} q_i q_j / \|x_i - x_j\|$.

This produces a layer-by-layer "physics encoding profile" for each architecture, revealing what is encoded, at what depth, and whether the tier of physics prior determines the profile.

Phase 2: Causal Verification

Probing reveals correlation, not causation. For each identified encoding, we perform causal ablation: project out the probed direction from the residual stream / hidden state, re-run inference, and measure the MSE increase. We additionally perform activation patching by corrupting a single particle's charge (flip $q_i \to -q_i$) and tracing which layers / components carry this change to the output trajectory. Together, these establish which physical representations are causally necessary for correct prediction.

Phase 3: Representational Geometry

Compute centered kernel alignment (CKA) between all pairs of architectures at matched depths to produce a representational similarity matrix. This reveals whether the two equivariant GNNs

(EGNN, Attention EGNN), the equivariant transformer, and the HNN develop structurally equivalent or divergent internal representations despite producing similar outputs.

Phase 4: Generalization Dynamics

Train all models across a range of dataset sizes with extended training budgets. Track probe accuracy for key physical quantities over training steps, alongside train and validation MSE, to test for grokking-like transitions. The hypothesis: for physics-explicit architectures, the emergence of correct physics representations and generalization co-occur; for symmetry-constrained models, geometry (distance, direction) is learned early but force magnitude may never be explicitly encoded.

Significance

This work makes contributions at the intersection of two active research areas: physics-informed machine learning and mechanistic interpretability.

For physics ML: The field currently evaluates models by output accuracy and symmetry preservation. This thesis introduces a third axis of evaluation, *representational alignment with physical law*, and provides the first systematic comparison of physics-explicit vs. symmetry-constrained vs. unconstrained architectures on this axis. The results will inform the principled selection of architecture type for scientific ML tasks where interpretability and trustworthiness matter.

For interpretability: The overwhelming majority of mechanistic interpretability work targets transformer language models. Physics simulation models are an underexplored domain with a critical advantage: the ground-truth computation is analytically known. This makes it possible to ask precise, falsifiable questions ("does layer 3 encode Coulomb potential?") rather than open-ended ones ("what does this attention head do?"). Insights about representation formation in physics models may generalize to other structured scientific domains.

The broader question: Does baking in known symmetries cause a model to internalize the corresponding physics, or merely constrain its output transformations while leaving internal computation opaque? This distinction matters practically: a model that has internalized Coulomb's law should generalize to novel charge configurations and particle counts; one that has learned a correlated shortcut may not.

References

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